HealthSense: Unobtrusive Continuous Stress Monitoring Using a Novel Dual ECG-PPG Patch

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Overview: HealthSense, a novel flexible wearable device that integrates ECG, PPG, and IMU sensors to continuously monitor stress, demonstrating high user comfort and effective stress prediction through machine learning models.

Introduction

Stress is a widespread global health issue, affecting over 60% of adults and leading to various physiological and psychological problems. Traditional stress monitoring methods (self-reports, cortisol measurements, blood pressure checks) have limitations in accuracy, convenience, and practicality for continuous monitoring. Researchers are developing continuous monitoring solutions using wireless and mobile sensors, but existing devices often have issues with skin conformity, comfort, or lack the ability to measure both ECG and PPG simultaneously.

We introduce HealthSense, a novel flexible and skin-conformable device that integrates ECG, PPG, and IMU sensors into a single wearable. A stress-induction study with 11 participants was conducted to assess the comfort of wearing HealthSense and the feasibility of stress prediction.

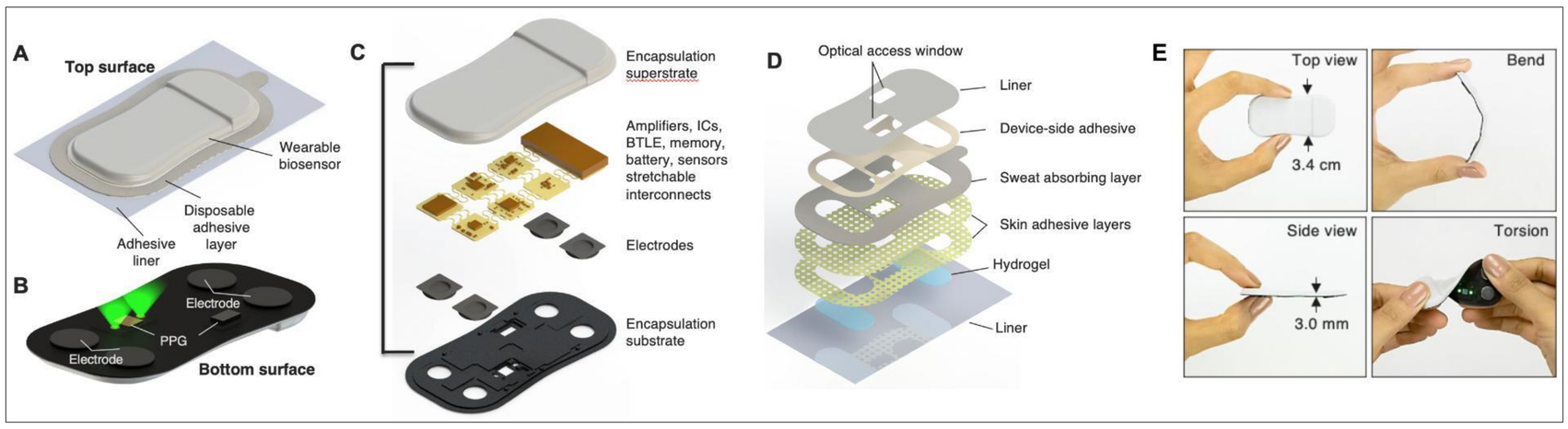
Results

The HealthSense device demonstrated high user acceptability, with all participants rating it as comfortable (≥4 on a 5-point Likert scale) and over 80% willing to wear it for extended periods.

In terms of stress prediction, the Gradient Boost Machine (GBM) model outperformed other models, achieving high precision, recall, and F1 scores for both perceived stress (0.855/0.859/0.855) and physiological stress (0.840/0.738/0.877) respectively.

The SHAP analysis revealed that pulse arrival time (PAT), systolic time, and pulse were the most influential features in predicting physiological stress, with systolic time showing an inverse relationship to stress levels and pulse rate positively correlating with increased stress.

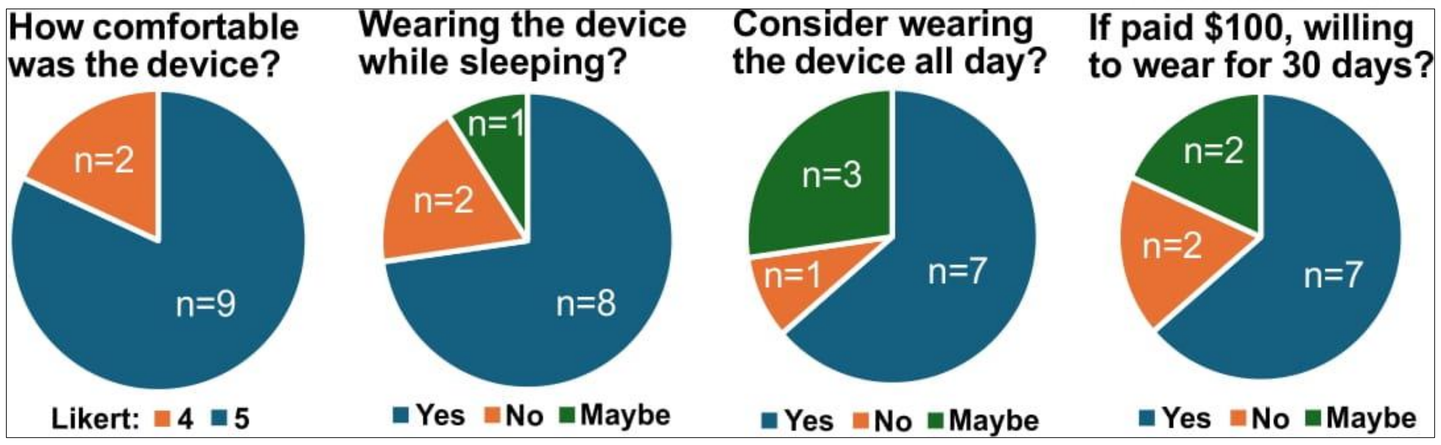
Methods



Device Design: a compact, wearable patch with 24-hour skin-friendly adhesion. It features ECG, PPG, and IMU sensors, with a battery life of up to three days.

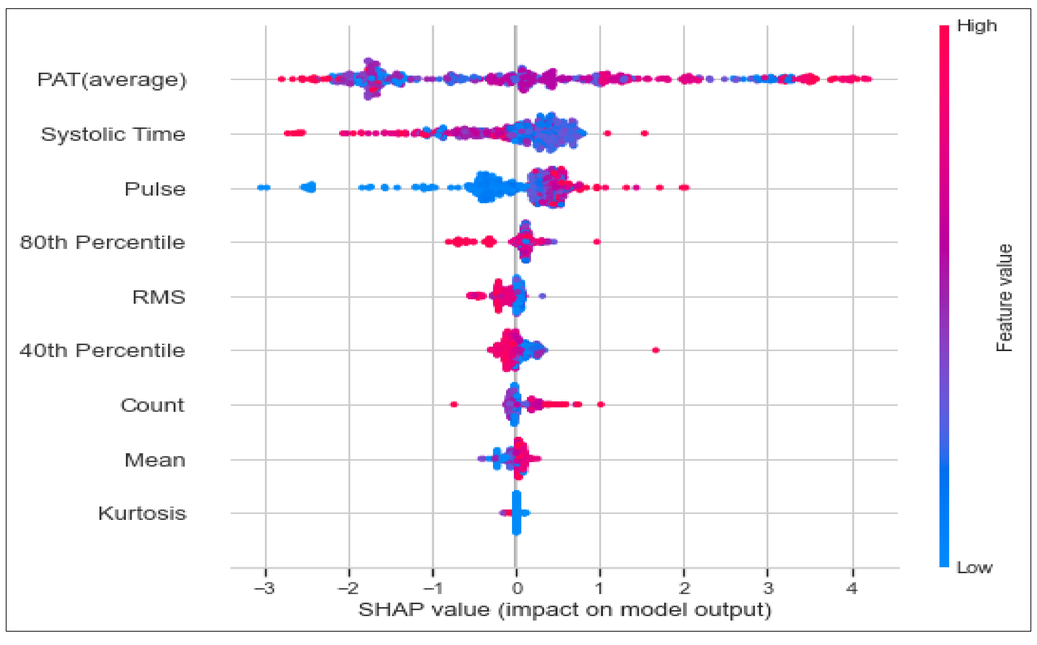
User Study: We conducted a stress induction study with 11 female participants aged 24 to 64. The study involved a series of activities designed to induce stress (PASAT, video gaming, etc). Participants wore the HealthSense device during the activities. After each activity, they completed a micro-EMA (Ecological Momentary Assessment) for their stress levels.

Wearability Assessment: At the end of the lab session, participants completed a survey about the comfort and feasibility of wearing the device in real-world settings.



Stress Prediction: 4 machine learning models were evaluated to predict perceived and physiological stress by features extracted from the ECG and PPG signals. Segmentation (1-min) and noise filtering were applied to isolate clean signals.

Model	Perceived Stress			Physiological Stress		
	Precision	Recall	F-1	Precision	Recall	F-1
GBM	0.855	0.859	0.855	0.840	0.738	0.877
DT	0.809	0.806	0.805	0.756	0.727	0.836
ADA	0.683	0.747	0.695	0.814	0.649	0.822
RDF	0.804	0.797	0.766	0.800	0.500	0.783



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